

MKM-EF-001 — Analysis

Component: Event Frequency Model — MKM-EF-001 (governance record v1.1.0) **Scope:** Empirical analysis of the model's measured behaviour on the halong catchment — the reprice it produces, the population-coverage correction, the dispersion regime, and the defects surfaced and fixed during construction. **Date:** 2026-07-28.

1. The problem the model corrects

The platform previously priced PRS off $\text{flood_count} / \text{num_storms}$ — a conditional probability with no time dimension, equivalent to setting the event arrival rate $\lambda = 1$ (“one storm = one year”). The Event Frequency Model supplies the missing rate: it counts *events* (storm sequences inside the 168-hour hours clause) and annualises the per-event flood conditional as $1 - \exp(-\lambda \cdot p)$.

2. Measured reprice

On halong the annualisation reprices **downward, 0.35–0.56×** the legacy metric. The severe-flood return period moves from the pre-frequency 8-10 years to 15-22 years; a representative property spread moves ~1100 to ~590 bps. The direction is not obvious a priori — during development it reversed twice as defects were found — so the figure is quoted only against the settled mechanism.

3. Population-coverage correction

The stress catalogue MKM-SS-001 oversamples severe events (it exists to train the classifier) and holds no minimal/baseline events at all, though those carry ~73% of the population mass. Naively resampling it answers $P(\text{flood} \mid \text{event is at least moderate})$, not $P(\text{flood} \mid \text{event})$. The model reweights the catalogue with population weights (8% severe-or-worse, owner judgement) and scales the conditional by the catalogue **coverage**, so events absent from the catalogue count at zero rather than being renormalised onto the severe tail. Left uncorrected, a configured 8% severe share read as an effective 29.6% — a 3.7× overstatement.

4. Dispersion regime

The per-gauge chi-square dispersion test finds halong's annual event counts **under-dispersed** (dispersion index ~0.46–0.97). No family on the Poisson–Negative-Binomial axis represents under-dispersion below the Poisson variance floor, and on synthetic data it is a declustering artefact rather than a physical regularity, so it is flagged and Poisson is selected as the nearest fittable family. The distributional form is in any case a second-order sensitivity (see the sensitivity analysis).

5. Defects surfaced and fixed

Building the model exposed five pre-existing platform defects, each corrected:

1. **Unseeded gauge response model** — every hazard curve the platform produced was irreproducible at $\pm 40\%$; now seeded per gauge and per (gauge, storm).
2. **Catalogue coverage** (as §3) — the 3.7× tail overstatement.
3. **Two field-naming traps** — `flood_events[].storm_id` holds *sequence* ids; `num_storms` was actually the sequence count. Both produced confident wrong answers.
4. **A units error in the basis leg** — an annual probability multiplied by an event count produced transmission rates near 200% (bounded by 1).
5. **Five missing BRI ratings** on commercial assets.

6. What the analysis does not establish

The measured reprice rests on the **unvalidated λ** . On a synthetic catchment the gauge record is generated from an assumed frequency, so the model recovers its own input; the per-gauge extraction validates the extraction code, not the rate. The λ sensitivity analysis shows the book inherits λ 's uncertainty roughly one-for-one, so the reprice magnitude is only as trustworthy as λ 's level — which requires real gauge data to validate (plan §5).

Reproduce: `python app.py port --halong --all` then the `propertyhc` / `gaugehc` outputs; measured figures from the `halong` build of 2026-07-28.